

FPGA-Based Multi-Sensor Monitoring System with MicroBlaze Processor

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Abstract— This project implements a comprehensive environment monitoring system utilizing FPGA technology. Developed on the Digilent Nexys A7-100T platform with a Xilinx Artix-7 FPGA, the system integrates multiple sensor modules which are the ambient light sensor, temperature sensor, and maxsonar sensor. A MicroBlaze soft processor orchestrates data acquisition from the Pmod peripheral modules while driving an RGB OLED display for real-time visualization. The design exemplifies embedded systems engineering principles through its modular architecture, hardware-software co-design methodology, and systematic validation approach.

Group Member Contributions	
Jad Al Jourdi	-Created block designs for all PMods individually and combined on Vivado. -Made vitis workspaces for all PMods individually and combined (OLEDRgb + ALS + MAXSONAR) -Documentation: Proofreading and Editing
Raihan Uddin	Implementation of TMP3 and it's pin mapping
Raphael Catchpole	Documentation: concept, creation, formatting, drafting

I. PROJECT OVERVIEW

A. Key Features

The system integrates multiple environmental sensors measuring and monitoring ambient light, temperature and distance allowing for comprehensive collection of data. The real-time visualization is achieved through an OLED. A USB-UART interface is used for parallel Communication, debugging with a PC, as well as flashing the FPGA hardware

Hardware architecture is modular and supports incremental sensor integration and verification through the use of the predefined Pmod IP blocks within Vivado. FPGA bitstreams are used for deployment across different Hardware configurations, allocating the proper hardware resources required within the FPGA.

B. Development Platforms

The Target board is Digilent Nexys A7 using a Microblaze processor. The hardware was designed in Xilinx Vivado 2021.1 using IP blocks and Vitis 2021.1 was used for writing the embedded software in C.

II. HARDWARE PLATFORM

A. Target Development Board: Digilent Nexys A7-100T

The system is implemented on the Digilent Nexys A7-100T FPGA development board, featuring:

- FPGA: Xilinx Nexys A7-100t
- Clock System: 100MHz crystal oscillator
- Memory: 256MB DDR2, 16MB Quad-SPI Flash
- I/O Interfaces:
 - 4 Pmod connectors (JA, JB, JC, JD)
 - USB-UART bridge for serial communication
- Reset: Dedicated CPU reset button

B. Hardware Architecture

The system centers around a MicroBlaze soft processor configured with 128KB of local memory for program and data storage. An AXI4 interconnect facilitates communication between the processor and peripheral interfaces, and a USB-UART interface for serial communication.

- Processor: 32-bit MicroBlaze at 100MHz
- Memory: 128KB local Block RAM (BRAM) for program and data storage
- Bus Architecture: AXI4 interconnect for peripheral communication
- Debug Interface: MicroBlaze Debug Module (MDM) for JTAG programming

C. Peripheral Interfaces

Sensor interfaces employ standard communication protocols appropriate to each peripheral: SPI for the ambient light sensor (PmodALS), I²C for temperature sensing (PmodTMP3), and UART for ultrasonic distance measurement (PmodMAXSONAR). This protocol diversity demonstrates the system's capability to interface with various sensor types while maintaining a consistent software architecture.

- *Pmod JA: ALS*
- *Pmod JB: RGB OLED display interface (PmodOLEDrgb)*
- *Pmod JC: TMP3*
- *Pmod JD: MAXSONAR*
- *USB-UART: Serial communication for data logging and debugging*

D. Clock and Reset

Clock management utilizes a clock wizard to generate the 100MHz system clock from the board's oscillator, with proper reset synchronization ensuring reliable startup and operation. The clocking wizard also provides any additional clocks at different frequencies for the peripherals that require any different clock frequency. The physical implementation adheres to the Nexys A7-100T's pin constraints, with careful attention to signal integrity and timing closure.

- *Clock Wizard*: Configurable clock generation from 100MHz input
- *Reset Controller*: Synchronized reset distribution across clock domains

III. SYSTEM ARCHITECTURE

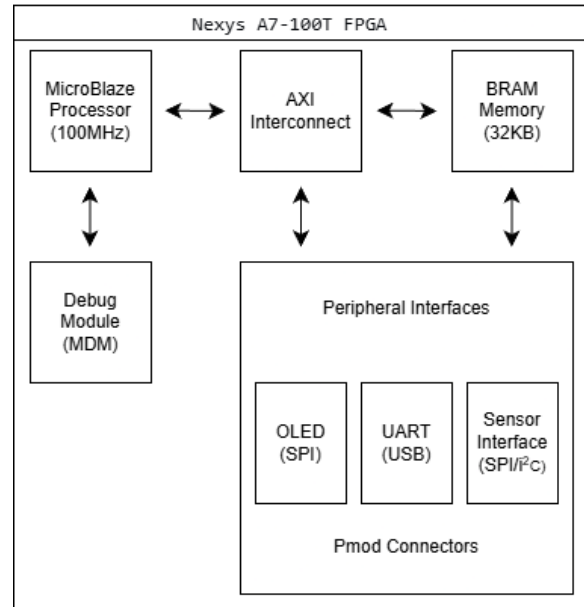


Fig. 1. Block Diagram Overview

A. Sensor Interfaces

1) Ambient Light Sensor (PmodALS):

- Interface: SPI communication protocol
- Sensor: Texas Instruments OPT3001
- Range: 0.01 to 83,000 lux
- Integration: Custom IP block or AXI-SPI controller

2) Temperature Sensor (PmodTMP3):

- Interface: I²C communication protocol
- Sensor: Texas Instruments TMP100
- Range: -55°C to +125°C
- Accuracy: $\pm 2^\circ\text{C}$ (max) from -25°C to +85°C

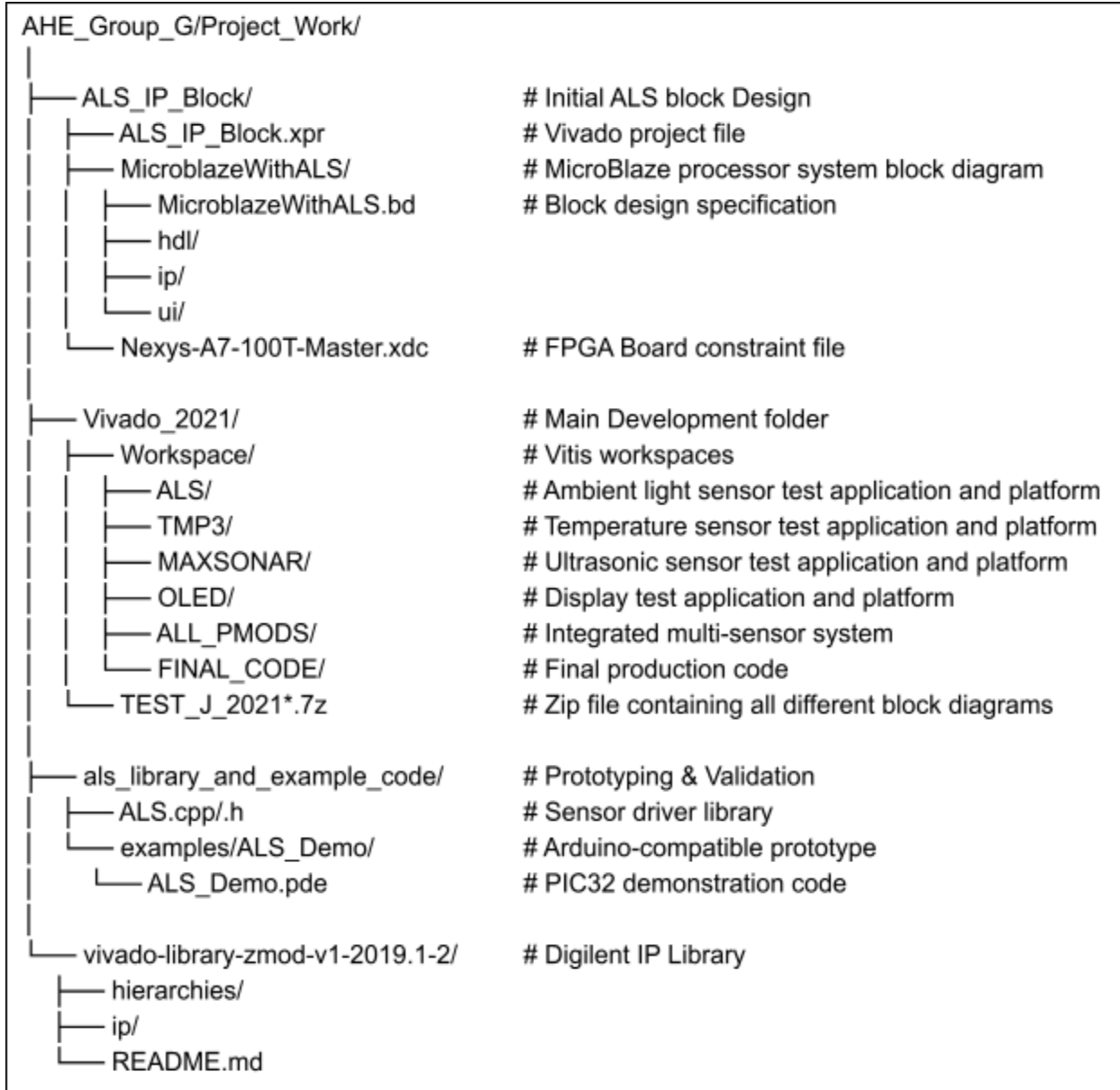
3) Ultrasonic Distance Sensor (PmodMAXSONAR)

- Interface: UART serial communication
- Sensor: MaxBotix MB1000
- Range: 20cm to 765cm
- Resolution: 1cm

4) OLED Display (PmodOLEDrgb)

- Interface: SPI communication
- Display: 96x64 RGB OLED
- Controller: Solomon Systech SSD1331

IV. REPOSITORY STRUCTURE



A. Key Files Explanation

- .xpr: Vivado project file containing hardware design configuration
- .bd: Block Design file defining system interconnect and components
- .xdc: Constraints file mapping logical ports to physical FPGA pins
- .xsa: Xilinx Support Archive for hardware-software interface

V. DEVELOPMENT METHODOLOGY

The project followed a phased development methodology beginning with requirements analysis and sensor selection. After the sensors were chosen, deployment of their respective IPs within Vivado was necessary with the use of a MicroBlaze soft core processor. All PMods went through the development methodology individually before being integrated with one another at the final run-through. During the first phase within Vivado, all the interfaces had to be defined within the block diagram and the respective constraints had to be configured accordingly.

After the hardware bitstream is successfully generated, it must be exported to be able to be imported into a new Vitis Workspace. At this phase the OS and hardware are automatically configured during the export into Vitis, and the platform is generated accordingly. After building the platform successfully an application component must be created. Here is where the main.c file resides and must be edited according to the intended functionality. (Example code can be found for the PMods within the platform component when using an IP). After which the application must be built (The platform must always be built first before the application).

The final phase of the development methodology is simply running the application on the intended hardware. At this phase testing of the written logic is necessary to ensure the hardware works as intended. The Github repository reflects this methodological approach. All Pmods went through these phases before being combined into a single block diagram which then is exported into Vitis and follows the development methodology to the final step.

A) Phase 1: Vivado – Hardware Bitstream Generation

- Developed Vivado project targeting the Nexys A7-100T FPGA.
- Integrated a MicroBlaze processor system with memory, AXI interconnect, and clocking infrastructure.
- Added Digilent IP cores for Pmod interfaces
- Generated the hardware bitstream with configured peripherals.

B) Phase 2: Vitis – C Application Development

- Created standalone applications in the Vitis workspace.
- Developed C code using the hardware driver example code provided for each sensor (ALS, TMP3, MAXSONAR) and the OLED display.
- Implemented data acquisition, real-time display output, serial logging, and calibration routines.

C) Phase 3: Hardware Deployment & Testing

- Loaded the generated bitstream onto the FPGA.
- Ran integrated C application for multi-sensor operation.
- Verified real-time data acquisition (< 100ms latency), display functionality, and serial communication.
- Finalized with full-system validation and documentation.

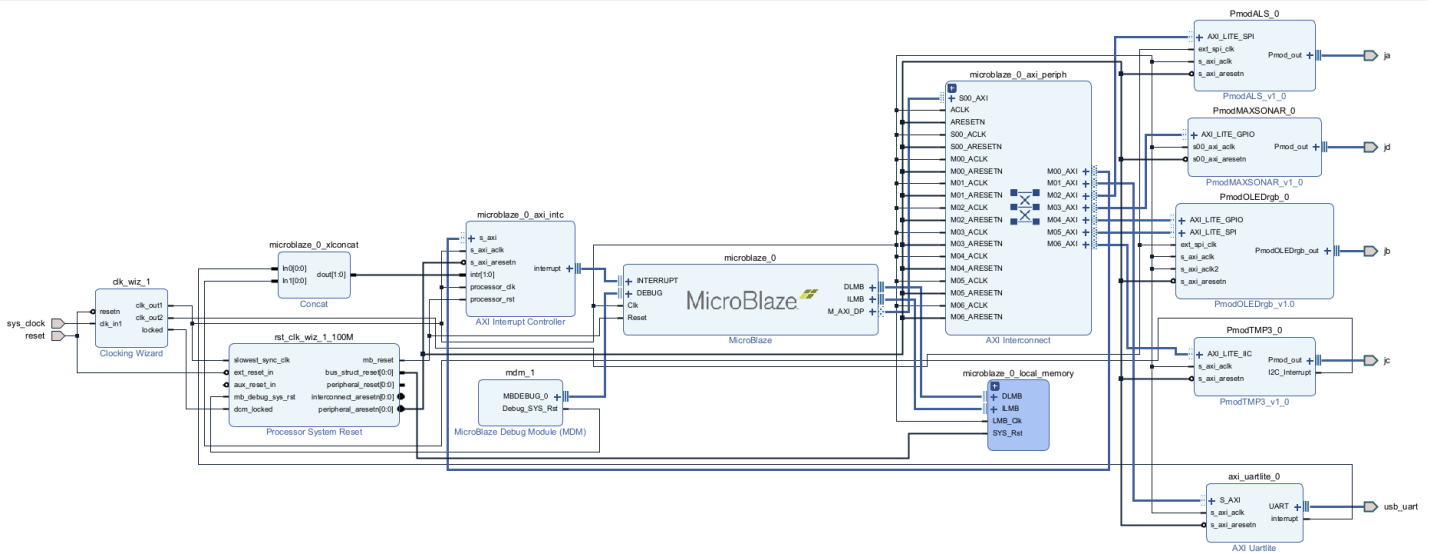


Fig. 2. MicroBlaze-based embedded system showing processor, AXI interconnect, peripheral modules, and control components.

```

10 //*****
11 * main.c - MAXSONAR + ALS -> OLED update every 1 second (MicroBlaze)
12 * - Shows distance in centimeters
13 * - Shows light value
14 *
15 //*****
16
17 #include "PmodALS.h"
18 #include "PmodMAXSONAR.h"
19 #include "PmodOLEDRgb.h"
20
21 #include "sleep.h"
22 #include "xil_cache.h"
23 #include "xil_printf.h"
24 #include "xparameters.h"
25 #include "xil_types.h"
26
27 #include <stdio.h> // sprintf
28
29 //***** Macro Definitions *****/
30
31 #define PMOD_MAXSONAR_BASEADDR XPAR_PMODMAXSONAR_0_AXI_LITE_GPIO_BASEADDR
32
33 #ifndef _MICROBLAZE_
34 #define CLK_FREQ XPAR_CPU_M_AXI_DP_FREQ_HZ
35 #else
36 #define CLK_FREQ 100000000
37 #endif
38
39 #define ALS_SPI_BASEADDR XPAR_PMODALS_0_AXI_LITE_SPI_BASEADDR
40 #define OLED_GPIO_BASEADDR XPAR_PMODOLEDRGB_0_AXI_LITE_GPIO_BASEADDR
41 #define OLED_SPI_BASEADDR XPAR_PMODOLEDRGB_0_AXI_LITE_SPI_BASEADDR
42
43 #define UPDATE_PERIOD_US 1000000 // 1 second
44
45 //***** Global Variables *****/
46
47 PmodMAXSONAR sonar;
48 PmodALS als;
49 PmodOLEDRgb oled;
50
51 //***** Cache Helpers *****/
52
53 static void EnableCaches() {
54     #ifndef _MICROBLAZE_
55     #if XPAR_MICROBLAZE_USE_ICACHE
56     Xil_IcCacheEnable();
57     #endif
58     #if XPAR_MICROBLAZE_USE_DCACHE
59     Xil_DcCacheEnable();
60     #endif
61 }
62
63 static void DisableCaches() {
64     #ifndef _MICROBLAZE_
65     #if XPAR_MICROBLAZE_USE_DCACHE
66     Xil_DcCacheDisable();
67     #endif
68     #if XPAR_MICROBLAZE_USE_ICACHE
69     Xil_IcCacheDisable();
70     #endif
71 }

```

Fig. 3. Vitis IDE showing ALL_PMODS main.c str

VI. IMPLEMENTATION DETAILS

A. Vivado Hardware Design Process

1) Project Creation:

- Created new Vivado project targeting Nexys A7-100T
- Imported Digilent IP library for Pmod support
- Applied board constraint file (Nexys-A7-100T-Master.xdc)

2) Block Design Development:

- Added MicroBlaze processor with 128KB local memory
- Configured AXI interconnect for peripheral access
- Added AXI UART Lite for USB communication
- Configured clock wizard for 100MHz system clock

3) Constraint Management:

- Mapped `ja_pin*_io` ports to Pmod JA connector, and respectively to JB, JC, and JD connectors
- Assigned `sys_clock` to 100MHz oscillator pin
- Connected reset to CPU reset button
- Configured USB-UART interface pins

4) Synthesis & Implementation

- Ran synthesis to generate netlist
- Performed place-and-route for physical implementation
- Generated bitstream for FPGA configuration

B. Vitis Software Development Process

1) Platform Creation:

- Exported hardware platform as `.xsa` file
- Created Vitis workspace with platform project
- Configured standalone operating system

2) Application Development:

- Created separate applications for each sensor
- Developed main application loop for data acquisition
- Implemented sensor-specific driver functions
- Added display update routines
- Integrated serial communication for data output

3) Build & Debug

- Compiled applications for MicroBlaze processor
- Used JTAG debugging via MicroBlaze Debug Module
- Validated functionality with hardware-in-the-loop testing

C. Key Technical Challenges & Solutions

1) Memory Constraints:

- Challenge: Limited BRAM for program and data storage
- Solution: Optimized code size, implemented efficient data structures

2) Sensor Synchronization:

- Challenge: Meeting 100MHz timing requirements with multiple peripherals
- Solution: Using the clocking wizard within the block diagram to output the required clock frequencies.

3) Display Refresh rate:

- Challenge: Coordinating multiple sensors with different interface speeds
- Solution: Implemented state machine for sequential sensor polling

4) Version Control:

- Challenge: Some members using the Vitis Classic IDE and others the Unified IDE
- Solution: Reverting some member to a classic IDE in order to be able to import a `.zip` file that was exported through vitis.